UMJI FA22 VG101 Homework 8

Assigned: 29 Nov. 2022 Due: 6 Nov. 2022 23:59pm

Notes for Submission

Please submit to the JOJ for grading purpose (**follow the naming rule on JOJ**).

For backup purpose, you should also submit your homework to the Canvas. For canvas submission, please name the file using the format: `hw8-[Your Last Name]-[Your First Name]-[Your SJTU ID].c` For example, `hw8-Yang-Zijun-521370910109.c`

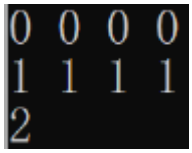
Homework Problem

Complete the class `MySpacetime` to represent the spacetime event (x, y, z, ct) in the provided HW8.cpp so the following main function can be run. Please do NOT change the provided main function.

```
int main() {
    double x, y, z, ct;
    cin >> x >> y >> z >> ct;  // input the coordinates of the first spacetime
event
    MySpacetime st1(x, y, z, ct);
    cin >> x >> y >> z >> ct;  // input the coordinates of the second spacetime
event
    MySpacetime st2(x, y, z, ct);
    cout << st1.Interval(st2);  // output the interval
    return 0;
}
```

Here the member function `Interval` will return the interval from `this` spacetime event to the other spacetime event, which is defined as $(x2-x1)^2+(y2-y1)^2+(z2-z1)^2-(ct2-ct1)^2$

where (x1, y1, z1, ct1) and (x2, y2, z2, ct2) are the spacetime coordinates of the two events. For example, suppose st1, st2 are `MySpacetime` objects, then `st1.Interval(st2)` will return the interval between st1 and st2. The following is an example run of the program:


0 0 0 0
1 1 1 1
2